Quantifying and Attributing Polarization to Annotator Groups

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Abstract

Current annotation agreement metrics are not well-suited for inter-group analysis, are sensitive to group size imbalances and restricted to single-annotation settings. These restrictions render them insufficient for many subjective tasks such as toxicity and hate-speech detection. For this reason, we introduce a chance-corrected metric that attributes polarization to various annotator groups. Our metric enables direct comparisons between heavily imbalanced sociodemographic and ideological subgroups across different datasets and tasks, while also enabling analysis on multi-label settings. We apply this metric to three datasets on hate speech, and one on toxicity detection, discovering that: (1) Polarization is strongly and persistently attributed to annotator race, especially on the hate speech task. (2) Religious annotators do not fundamentally disagree with each other, but do with other annotators, a trend that is gradually diminished and then reversed with irreligious annotators. (3) Less educated annotators are more subjective, while educated ones tend to broadly agree more between themselves. Finally, we estimate the minimum number of annotators needed to obtain robust results, and provide an open-source python library that implements our metric.

Keywords: Polarization, annotation, annotators, socio-demographic, statistics, metric

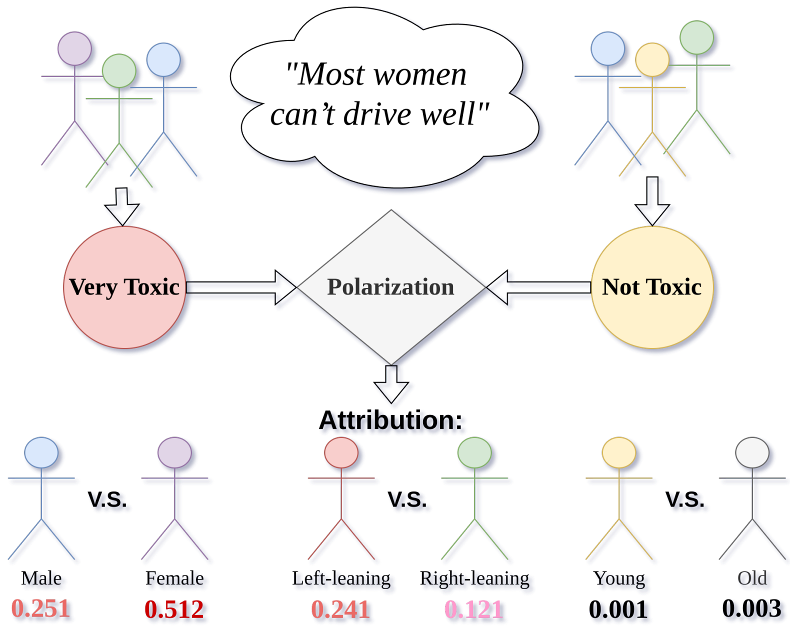

Fig. 1 The *apunim* framework attributes annotator polarization to various annotator subgroups.

1 Introduction

Annotations are essential for a wide range of tasks, especially in fields with subjective judgements, such as content moderation, toxicity detection and hate speech detection. These tasks are especially important for training supervised learning models that remove or flag comments targeting vulnerable and minority groups in online spaces, where they are often vulnerable [1, 2]. However, annotations for such tasks are subjective and typically aggregated based on majority-voting [3–5]. This may at best limit the usefulness of the data, or at worst lead to biased and misconfigured systems [6].

The core of the problem is simple; if a comment targets marginalized groups, there can be cases where most annotators overlook it, while the few marginalized annotators vehemently disagree with the majority. These fundamental disagreements are then ignored via aggregation (e.g., majority label voting) or filtering (e.g., keeping only items with large overall annotator agreement). Racist comments that are presented with coded language (usually referred to as "dog-whistles" [7]) are a common example; these comments may not be picked up by most annotators, but only by ones belonging in the targeted minority. Furthermore, majority-group annotators may themselves be biased against minority groups as was the case for toxicity models disproportionally marking African American speech as "toxic" [6].

Inter-annotator agreement is often used to detect such cases [5]. Metrics based on agreement mostly measure annotation variance, which may not be an optimal problem formulation for detecting such cases, and fundamentally constrain analysis on single-label annotation. *Polarization* is a better instrument in this regard, since it formulates annotation analysis as a cluster identification problem [8, 9]. While we can detect polarization [8, 9], there is little work on how to detect whether it is actually caused by marginalized groups. An existing mechanism operating on individual comments [9], operates only as a binary classifier, and as such does not produce quantifiable/comparable results. It also only tests for the extreme case where *all* present polarization can be attributed to any single annotator group, which is extremely rare in real-life datasets (see Appendix B.2).

We instead propose the *apunim ("Aposteriori Unimodality") metric*. Unlike previous methods which provide only a binary decision (yes/no attribution) [4, 9], *apunim* quantifies the level of attribution for each annotator subgroup. Additionally, unlike previous approaches which only work on a comment-to-comment basis, *apunim* generalizes results for all annotated comments (or "items", since the metric is generalizable to many domains), which is particularly important for our formulation due to the typically limited number of annotations per item. By considering the entire set of items, *apunim* helps to identify genuine polarization and distinguish it from chance attribution, which is essential when analyzing large datasets. Our contributions are:

1. We introduce a new metric that quantifies polarization, which can be attributed to specific annotator groups, and we develop a parametric test for statistical significance. This metric works for both single-label and multi-label annotation setups, is generalizable across any domain and modality (e.g., video/image classification, toxicity/hate-speech detection) and enables quantitative comparisons across datasets and dimensions (e.g., gender, race, political affiliation).
2. We discover issues not yet acknowledged in literature (§3). We are the first to report the existence of *inherent polarization*: most annotated items having some degree of polarization that can not be attributed to *any possible partition* of annotators. We also find that annotator groups may exhibit opposing polarization attribution between items, which can cancel each other out when not handled properly.
3. We apply our metric to four Natural Language Processing (NLP) datasets focused on Toxicity and Hate Speech detection (§5). We find that annotator race/ethnicity contributes to polarization across all datasets, but more so on Hate Speech tasks. We also discover that less educated annotators are generally more subjective, and that in some cases, polarization is either concentrated around similar clusters of groups (e.g., age and education) or that annotators become more polarized the more extreme their opinions become (e.g., religion).

We release an open source Python library (see Appendix C) that implements the calculation of the metric and its statistical significance, which can be directly installed via PyPi.[1] Furthermore, we provide an estimation on the minimum number of annotators required for robust estimates (§6), and we release the code for the experiments, graph creation, and analysis on our project's repository.[2] Finally, we outline a few possible applications of our framework in §8.

[1] Library: https://pypi.org/project/apunim Documentation: https://dimits-ts.github.io/apunim
[2] Replication code: https://github.com/dimits-ts/aposteriori-unimodality

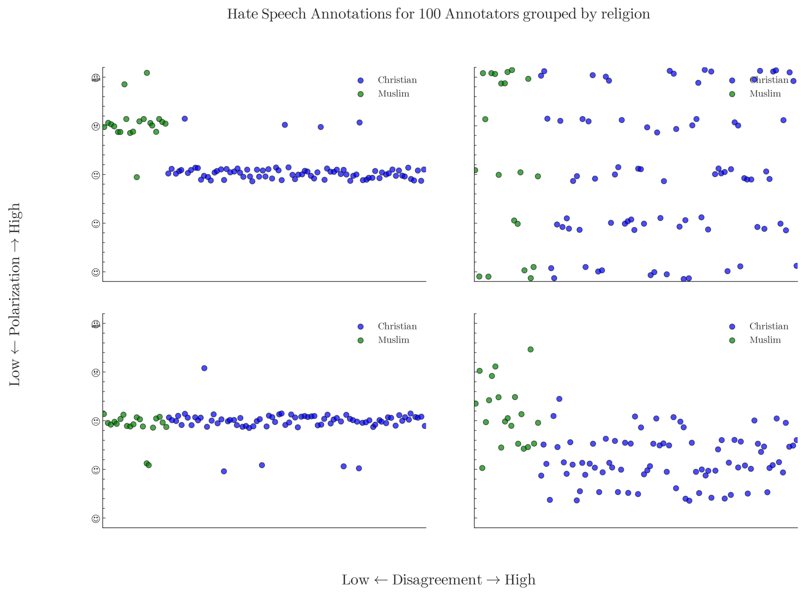

Fig. 2 Disagreement is similar to variance, overlooking cases where annotators may be "split down the middle". Polarization on the other hand, identifies multiple clusters even if they are close together.

2 Related Work

Annotator agreement

There are many popular metrics for measuring inter-annotator agreement such as Cohen's kappa, Krippendorff's alpha and Fleiss's kappa. All of these metrics attempt to distinguish genuine disagreement among annotators from noise (or "random disagreement") using statistical methods. However, most can not be generally used for comparing intergroup annotation agreement without extending them (e.g., computing pairwise Cohen's kappa for all groups), or placing constraints on when they can be used (e.g., no missing labels and same number of annotations per annotator for Fleiss's kappa). They have also been criticized for ignoring fundamental differences that can arise from different social, cultural and demographic backgrounds [10, 11], and for their inability to quantify and compare results across experiments [12].

Recent work has attempted to work around these limitations, for example by adapting Cohen's kappa for subgroup agreement [12] or creating a Bayesian model to explicitly differentiate between subgroup disagreement and noise [3]. While these extensions do improve the performance of these metrics, they are typically less explainable and intuitive. They are also constrained by the same basic assumptions of disagreement metrics, such as not being suitable for multi-label annotations (where annotators can pick more than one option for each item).

Annotator polarization

Due to their nature, disagreement metrics may not perform well when comparing annotator groups with greatly mismatched sizes [10, 11]. Figure 2 shows an example of an annotation task where the minority group annotators may be ignored when computing disagreement, due to the overwhelming number of majority-group annotations. Polarization [9] attempts to solve this problem by framing disagreement as a cluster identification problem, where similar annotations will belong in a single cluster, disregarding its size. This formulation, besides being appropriate for imbalanced groups, also enables metrics based on polarization to work with multi-label annotations, since we are no longer comparing individual annotations against those of other annotators, but rather each annotator's histograms (see §3.1).

Attributing polarization to subgroups is a relatively new idea. Under the name *aposteriori unimodality* [9], the original idea is that if polarization in a single item is positive, but zero for all annotator subgroups, then those subgroups are "causing" the exhibited polarization. Though promising, this idea is not practically usable, since subgroup polarization being exactly zero is an edge-case in real-life annotation tasks (see B.2). Additionally, getting polarization estimates from realistically few annotations per item is extremely noisy, which is especially problematic when the output is binary (i.e., an item either is or is not polarizing) instead of a quantifiable value. For example, when presented with noisy estimates of a metric in the [0, 1] range, we would ideally want to distinguish between cases where attribution is close to 0.001, and cases where attribution is 0.49.

Such a binary analysis has been applied to discover whether Large Language Models (LLMs) are biased in the same way as specific human subgroups [4]. This work shows that polarization is a versatile tool, but also highlights the constraints of using the non-quantifiable binary classification given by the traditional aposteriori unimodality framework. For instance, the authors have to resort to Cohen's kappa agreement for any kind of comparison between LLMs and human annotator groups, which downgrades polarization to the role of filtering texts, and as such their analysis is still constrained by the restrictions outlined above. The authors also showed how ordinal variables can extract "better" categorical labels via dataset-dependent thresholding, an idea that is promising yet limited (see §5.3).

3 How to attribute polarization

3.1 Problem Formulation

Annotator groups

Since our goal is to pinpoint which specific characteristics contribute to polarization, we need to split annotators according to their Personal Characteristics (PCs), which can be sociodemographic [6, 9, 13, 14], ideological or based on personal opinios and experiences [4, 15]. Each of these characteristics is a dimension Θ composed of multiple groups $\theta_1, \theta_2, \ldots$; e.g. if we are investigating gender, then $\Theta = \{\theta_{male}, \theta_{female}, \theta_{nonbinary}\}$.

Annotations

Let $d = \{c_1, c_2, \ldots\}$ be a dataset composed of multiple annotated items.[3] These items could be of any modality, such as comments in a discussion, images, videos or survey questions. We assume that how polarizing an item is depends on two variables: *inherent ("apriori") polarization*, and polarization caused by *different annotator PCs ("aposteriori" polarization)*. Inherent polarization refers to polarization that can not be explained by any arbitrary annotator grouping. It may be caused by the task itself, the instructions given to the participants, or pure chance. In previous work [9], it was assumed to be 0, although this is usually not the case (§3.4).

Each item c is assigned multiple annotators, which creates the annotation multiset $A(c) = \{(a_1, \theta_1), (a_2, \theta_2), \ldots\}$, where a is a single annotation. As an example, the annotations for a comment in a sentiment analysis task where we group the annotators by gender, would be formulated as:

$$A(\textit{"Could be better, could be worse"}) = \{(+, F), (-, M), (+, F), \ldots\}$$

Polarization

In the absence of polarization, annotation distributions are typically unimodal: annotators may disagree on finer details—shown as a bell-shaped distribution around a central mode—rather than on fundamentally different interpretations, which would instead give rise to multimodal distributions. To our knowledge, the only polarization metric explicitly grounded in this assumption is the normalized Distance From Unimodality (nDFU) [9]. The metric takes values in the [0, 1] range, with $nDFU \approx 0$ indicating no polarization (unimodal distributions) and $nDFU \approx 1$ corresponding to complete polarization (highly multimodal distributions). Our objective is therefore to quantify the extent to which the $(nDFU)$ score of a polarizing item can be (partially) explained by a given PC group θ.

3.2 Intuition

Polarization attribution

Intuitively, a group $\theta \in \Theta$ partially explains the polarization of an item c when the annotations of that group $A(c|\theta) = \{a(c; \theta') \in A(c) | \theta' = \theta\}$ exhibit higher polarization compared to the full set of annotations $A(c)$. Figure 3 exhibits a hypothetical example where a misogynistic comment is annotated for toxicity by male and female annotators. The annotations are generally polarized ($nDFU_{all} = 0.62$). The annotations by female annotators exhibit low polarization between themselves ($nDFU_{women} = 0.10$), since most agree that the comment is toxic. The set of male annotations, also shows low polarization ($nDFU_{men} = 0.37$), but for the opposite reason—most men agree that it is *not* toxic. This suggests that the overall polarization is driven by disagreements between male and female annotators.

Previous work [9] stopped on item-to-item (comment-to-comment) analysis. In practice however, this approach is inherently noisy due to a lack of information (i.e.,

[3] We use the general term "item" instead of domain-specific terms, even though the examples presented in the paper are comments. The mathematical term "point" is equivalent.

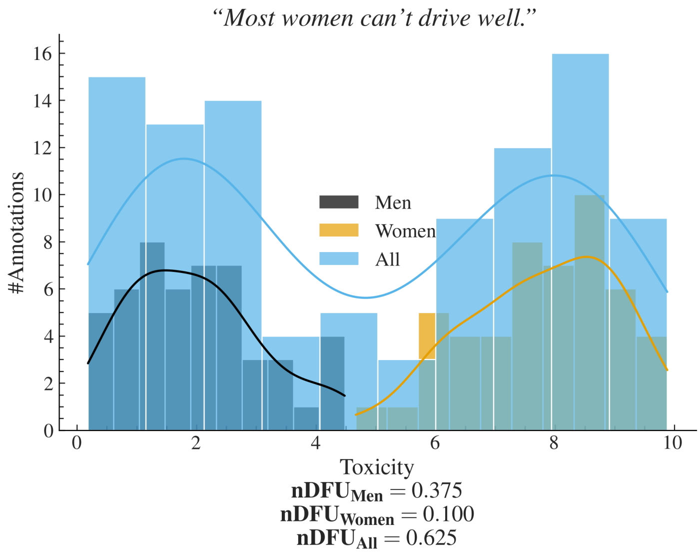

Fig. 3 Example of a polarizing comment, where male and female annotators agree between themselves, but disagree with the opposite gender.

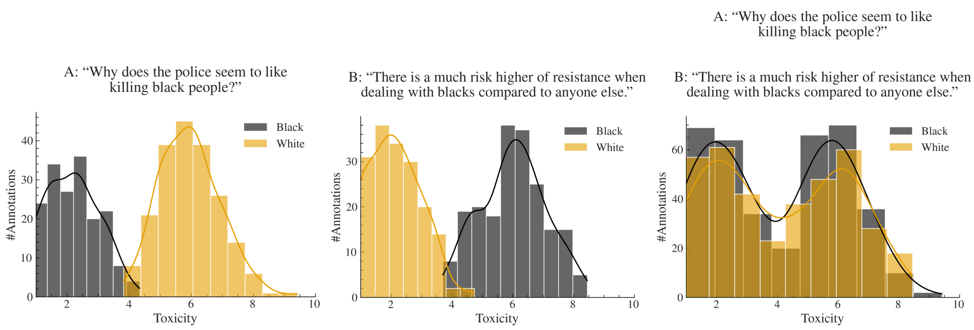

Fig. 4 Example of a polarizing discussion with two comments where annotators disagree about which one is toxic. When we aggregate the comments (right-most image), we unintentionally change the comparison: instead of comparing two unimodal distributions (each comment viewed separately) with two bimodal distributions (their aggregated annotations) as shown in the center and left-most images, we end up comparing two bimodal distributions formed by combining both comments within each group. This aggregation can hide where the polarization actually comes from.

limited annotations) relating to each individual item. In order to get meaningful information, we need to consider the annotations from all items, as the number of items is usually several orders of magnitude larger than the number of annotations per item (§4). We thus reformulate our hypothesis; if an annotator subgroup causes some of the underlying polarization in our dataset, we expect that this attribution will persist across multiple polarizing items.

Dataset polarization

Given this observation, we would be tempted to quantify dataset-level polarization attribution by simply aggregating all annotations in the dataset ($\cup_{c \in d} A(c)$) and comparing their polarization with that of $\cup_{c \in d} A(c|\theta)$.In fact this approach is very tempting since aggregating annotations solves the very real and important problem of annotation scarcity; each item may be annotated by only a handful of annotators (5 is a common number in many publications), leading to a best-case even split between groups (for 5 annotators, a best case scenario would be only two groups with a 3-2 split). Since nDFU is at its core a histogram-based estimation, three annotations may not be enough to acquire a reliable histogram, and thus estimate of polarization. This is discussed in more depth in §6.

However, arbitrarily combining annotations would not work well in practice. Figure 4 illustrates a hypothetical discussion with two comments, both of which are toxic, but where the two groups of annotators disagree on *which* comment is the toxic one. If we aggregate the annotations to a single discussion, the opposing polarization effects might balance each other out, leading to a false negative. This demonstrates that item-level polarization attribution has a *direction*, which is not captured by nDFU, and has to be taken into account when aggregating annotations between items.

3.3 Aposteriori polarization

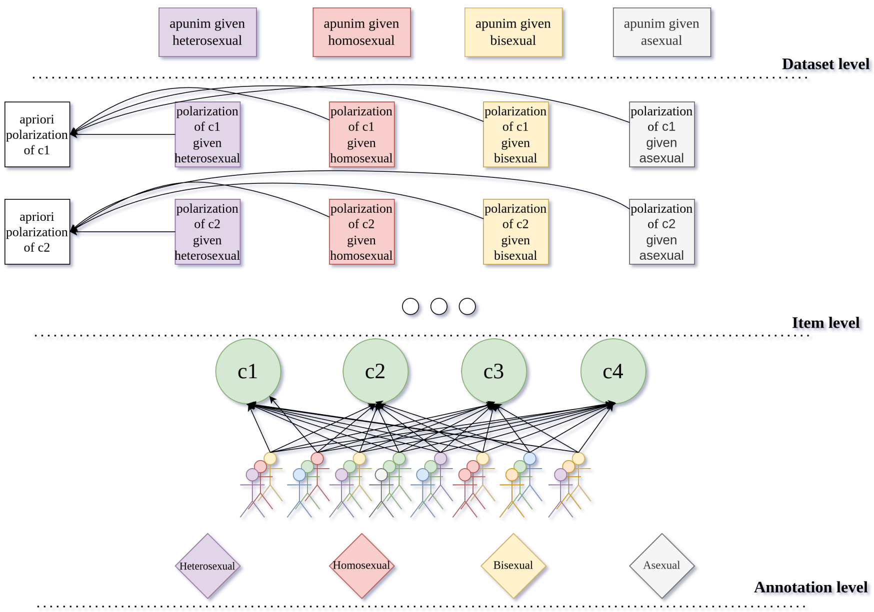

Fig. 5 An overview of how annotation information is sequentially aggregated for the *apunim* metric. For each item we calculate the apriori polarization and the sample aposteriori polarization given each PC group. We then aggregate both apriori and aposteriori polarization and compare them on the dataset level to obtain the polarization attribution ("apunim") values for each group.

We do not directly compare the differences in nDFU between $A(c)$ and $A(c \mid \theta)$, as in §3.2, to avoid comparing statistics between a subset and its superset. Standard practise dictates that we should compare the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this does not work in the (common) case where we are studying a PC with two subgroups.[4] If we applied the subset-vs-complement method in Figure 3 where we consider the gender dimension, we would end up subtracting the nDFUs of men and women, which are similar as they are both unimodal distributions, concluding that there is no polarization present.

3.3.1 Apriori Polarization

In a scenario in which each item was annotated by many annotators, we could use sampling or bootstrapping to compare apriori and aposteriori polarization per item. This, however, is an unrealistic assumption, since the costs of human annotation force researchers to use few annotators for each task [16]. To get around this limitation, we create random stratified splits that effectively "break up" the effect of any single isolated group on the polarization of the sample.[5] The randomly partitioned annotations are then sampled t times. Formally, the (average) apriori polarization for a single item c is then given by:

$$P_{apr}(c) = \frac{1}{t} \sum_{i=1}^{t} nDFU(\tilde{r}_i(A(c))) \tag{1}$$

where $\tilde{r}$ is the random partition operator with group sizes matching the group sizes of PC Θ. Unlike $A(c)$, we can use this formulation for direct comparisons with $A(c \mid \theta)$.

3.3.2 Dataset Polarization: Apriori and Observed

While the low number of annotators is an important limitation when acquiring the observed polarization [16], this is decisively not the case with the number of items themselves, which can range to up to hundreds of thousands (§4). Thus, we have the luxury of filtering them before computing our metric, as proposed in previous work [4]. Since our metric attributes polarization, unpolarizing items and those composed of only a single annotator group are useless. We define the subset S_d of dataset d such that:

$$S_d = \{c \in d \mid nDFU(c) > \alpha; \Theta^c > 1\}$$

where α is a small positive number and Θ^c is the number of distinct annotator groups in $A(c)$. This definition filters out a large fraction of unpolarized items (Appendix B.2), reducing noise and significantly lowering computational costs (Appendix D). Then, we can estimate how much polarization is generally present in the dataset, irrespective of

[4] Even when there are more than two groups, we may end up with no items that were annotated with more than one annotator of a minority group, meaning that we can not get a histogram estimation. This occurred frequently in datasets with low numbers of comments such as the Sap dataset (§5).

[5] For example, if we have 100 annotations, 80 of which are made by male annotators and 20 by female annotators, we will create random partitions (disregarding gender) of sizes 80 and 20.

any PC, by simply averaging the *apriori* polarization of its items:

$$P_{apr}^{d} = \frac{1}{|S_d|} \sum_{c \in S_d} P_{apr}(c) \tag{2}$$

Intuitively, a high apriori polarization score means that much of the polarization in the dataset is likely not driven by factors of the investigated Θ dimension (polarization insists disregarding that dimension). Similarly to Equation 2, we also define polarization that was *observed* by a specific group of annotators, e.g., θ, as:

$$P_{obs}^{d}(\theta) = \frac{1}{|S_d|} \sum_{c \in S_d} nDFU(A(c \mid \theta)) \tag{3}$$

This formulation implicitly assumes that dataset polarization can be estimated by averaging polarization scores across its items. We consider this assumption reasonable, because the mean is a simple and intuitive operator, exhibiting appropriate sensitivity to outliers. Given that polarization values lie in the [0, 1] range, a single item with extreme polarization is unlikely to substantially affect the estimate of overall polarization. In contrast, the presence of many such extreme values will meaningfully increase this estimate, which is the pattern we want to capture.

3.3.3 Apunim

We can now compare the polarization that is attributed to a single-dimensional annotator group (Equation 3) to a prior one (Eq. 2). We define our normalised metric, which we call as *"apunim" (Aposteriori Unimodality)*, as:

$$apunim(\theta) = \frac{P_{obs}^{d}(\theta) - P_{apr}^{d}}{1 - P_{apr}^{d}} \tag{4}$$

The scope of the statistic and its relationship with the previously presented statistics are demonstrated in Figure 5. Since $nDFU \to [0, 1] \Rightarrow apunim \to [-1, 1]$. The interpretation of this metric boils down to three scenarios:

- If $apunim(\theta) \approx 0$, the exhibited polarization among annotators of group θ does not meaningfully differ from all other groups. In other words, it is indistinguishable from noise (represented by the apriori polarization).
- If $apunim(\theta) > 0$, then the polarization observed in the annotation task rises when this group is added to the annotation pool (equivalently, intra-group polarization is lower than apriori polarization). For example, if we observe $apunim($*"African American"*$) = 0.3$, $apunim($*"White"*$) = 0.001$ in a hate-speech detection task, it is likely that African American annotators fundamentally disagree with white annotators on what constitutes racism, regarding dataset d.
- Conversely, if $apunim(\theta) < 0$, then the overall polarization *drops* when we add annotators from this group to this task. Mathematically, this occurs when intra-group polarization is higher than apriori polarization. This may indicate that *other*

Algorithm 1 P-value estimation for aposteriori polarization

Require: Dataset d, annotator group $\theta \in \Theta$
Require: Filtered items S_d
Require: Number of random partitions t
Ensure: p-value for $H_0 : apunim(d, \theta) \neq \overline{apunim}(d)$

$P^d_{obs}(\theta) = \frac{1}{|S_d|} \sum_{c \in S_d} nDFU(A(c \mid \theta))$ ▷ As in Equation 3
$P^d_{apr} = \frac{1}{|S_d|} \sum_{c \in S_d} \frac{1}{t} \sum_{i=1}^{t} nDFU(\tilde{r}_i(A(c)))$ ▷ As in Equation 2
$apunim = \frac{P^d_{obs}(\theta) - P^d_{apr}}{1 - P^d_{apr}}$ ▷ As in Equation 4
$rand\_apunims = \{\}$
for $i = 1, 2, \ldots, t$ **do**
  $rand\_obs = \frac{1}{|S_d|} \sum_{c \in S_d} nDFU(\tilde{r}_i(A(c))$ ▷ Single random partition
  $rand\_apriori = \frac{1}{|S_d|} \sum_{c \in S_d} \frac{1}{t} \sum_{i=1}^{t} nDFU(\tilde{r}_i(A(c)))$ ▷ Same as P^d_{apr}
  $rand\_apunim = \frac{rand\_obs - rand\_apriori}{1 - rand\_apriori}$
  $rand\_apunims = rand\_apunims \cup rand\_apunim$
end for
$p = Student\text{-}T(apunim \neq average(rand\_apunims))$
return p

dimensions (acting as confounding variables) *are exerting heavy influence on the sample* or that annotators of that group may be very polarized between themselves. For example, if we observe $apunim(Latino) = -0.3$, $apunim(White) = 0.001$ in a hate-speech detection task, it is likely that Latino annotators fundamentally disagree with *each other* (compared to African American or white annotators) on which texts in the dataset d constitutes racism.

3.4 P-value estimation

The strict condition $apunim(\theta) \approx 0$ is not sufficient due to inherent annotation variance. We initially formulated the following statistical hypothesis for any $\theta \in \Theta$:

$$H_0 : apunim(\theta) = 0 \text{ and } H_a : apunim(\theta) \neq 0$$

However, early experimentation showed that *apunim* values were rarely statistically 0, possibly due to inherent polarization (§3.3). This indicates that such a formulation is not strong enough, and would possibly lead to frequent Type I errors.

For this reason, we reformulate our hypothesis as follows; If the apunim value is statistically significant, we expect it to significantly differ from *apunim values exhibited by random annotator groups*. We use the individual randomized partitions $r_i(A(c))$ to compute pseudo-apunim values, which are then compared to the actual apunim value of the sample using a Student-T test [17], as shown in Algorithm 1. Thus, we test a much stricter hypothesis for any given $\theta \in \Theta$:

$$H_0 : apunim(\theta) = \overline{apunim}(d),\ H_a : apunim(d, \theta) \neq \overline{apunim}(d)$$

where $\bar{apunim}(d)$ denotes the mean apunim values of each random partition $\tilde{r}_i(A(c))$.

Since our test is parametric, it assumes that there exist (1) enough data-points to reliably run the means-test, and (2) enough annotations per PC group to estimate nDFUs in Equations 2-3. The former can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §4). We discuss the latter assumption in §6.

While the test operates on a single θ dimension in principle, in practice we often want to test polarization for all subgroups ($\forall \theta \in \Theta$). Since we are simultaneously considering $|\Theta|$ hypotheses, we apply a multiple comparison correction to the resulting p-values—specifically the Holms method [18]. The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses [19]). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

4 Datasets

Table 1 Overview of the datasets used in this study. *# Annotators* refers to the number of annotations per comment. For detailed information on the annotation counts for the comments of each dataset, see B.1.

Dataset	#Comments	#Annotators	#PC dims	Task
Kumar [15]	106,035	5	10	Toxicity
Sap [13]	626	5–6	3	Hate Speech
DICES-350 [14]	72,103	70–76	4	Hate Speech
DICES-990 [14]	43,050	123	4	Hate Speech

We use four datasets from current NLP research that include PC information at the level of individual annotators.[6] Specifically, we use the "Kumar" [15] and "Sap" [13] datasets, which are concerned with Toxicity and Hate Speech detection respectively, as well as the DICES-350 and DICES-990 datasets [14], which focus on LLM response safety. For the latter, we consider only labels relating to racism (`Q3_bias_overall`), thereby transforming the task to Hate Speech detection. While the Toxicity and Hate Speech detection tasks are not equivalent, they share substantial conceptual and operational overlap, as both require annotators to assess the presence and severity of harmful or abusive language. Thus, we expect the sources of annotator disagreement and bias that arise in these tasks will be broadly similar.

The datasets differ substantially in scale and annotation design. The Kumar dataset stands out for its large size, featuring more than 100,000 annotated comments, as well as 10 distinct PC dimensions. In contrast, the DICES datasets contain fewer annotated comments but involve more than an order of magnitude more annotators

[6] While we would like to test our framework on other research areas and domains, NLP remains the only large field to our knowledge that systematically investigates annotator biases and whose datasets comprise the necessary PC information to allow our study.

per comment than the other datasets. A brief overview of dataset characteristics is provided in Table 1. The Kumar and Sap datasets contain a large number of polarizing items, unlike the DICES datasets, where most items are unpolarized (Figure B2). This can most likely be attributed to the DICES datasets being composed of LLM-generated dialogues with already-aligned models tuned to avoid racist speech.

5 Results

5.1 Experimental Setup

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided PC dimensions can partially explain polarization in the toxicity/racism annotations, as shown in Table 1. We sample 20,000 comments from the Kumar dataset due to computational costs (Appendix D). Note that some groups are not included, either because the results were not statistically significant, or because there were no polarized items annotated by more than one annotator of these groups (see §3.3.2 about this restriction).

5.2 Which factors contribute to polarization?

Table 2: Statistically significant (****:$p < 0.01$, **:$p < 0.05$*) aposteriori unimodality results across all four datasets. High *apunim* values indicate that polarization can be attributed to this subgroup. Near-zero values indicate that polarization can not be attributed to the subgroup. Negative values indicate that there is more polarization among annotators of that subgroup compare to other subgroups (see §3.3). For the full results, consult Appendix E.2.

Dataset	PC dimension	Group	apunim	support
DICES-350	Race	African American	-0.0428***	9077
		Asian	0.0659***	8138
DICES-990	Age	Gen. X+	-0.0471***	14384
		Gen. Z	0.0644***	13741
	Education	High school or lower	-0.0751***	7419
	Gender	Man	0.0278***	32494
		Woman	-0.0235***	33471
	Race	African American	0.0909***	5810
		Latino	-0.1328***	6610
		Other	0.0753***	24321
		White	-0.1196***	11392
Sap	Ethnicity	Black	-0.2967***	1041
		White	0.1400***	2048
	Gender	Male	0.0942***	1635
		Female	-0.1810***	1333

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Table 2: (continued)

Dataset	PC dimension	Group	apunim	support
Kumar	Education	Master's degree	0.0147**	13420
		Bachelor's degree	0.0401***	34085
		Associate degree	-0.0113**	9122
		College, no degree	-0.0335***	16666
	Ethnicity	White	-0.0377***	45306
	Gender	Female	-0.0278***	40686
		Male	0.0275***	37929
	Has Been Targeted	False	-0.0633***	45234
		True	0.0349***	24491
	Is Parent	No	-0.0707***	37231
		Yes	0.0535***	41493
	Is Transgender	No	-0.0894***	11978
	Political Affiliation	Conservative	0.0331***	23010
		Independent	-0.0157***	22818
		Liberal	-0.0287***	32917
	Religion Important	No	-0.0804***	26008
		Not very	-0.0154***	10055
		Somewhat	0.0210***	20169
		Very	0.0487***	27284
	Seen Toxicity	False	0.0311***	20185
		True	-0.0717***	43265
	Sexual Orientation	Bisexual	0.0175***	10005
		Heterosexual	-0.0571***	38048
	Technology Impact	Somewhat positive	-0.0265***	40477
		Very positive	0.0208***	22471
	Toxicity Problem	Rarely	0.0177***	14457
		Frequently	-0.0217***	24774
		Very Frequently	-0.0145***	10656

Polarization is consistently attributed to annotator race/ethnicity across all datasets, and especially so on hate-speech detection tasks

Across all four datasets, we observe a consistent pattern in which annotation polarization can be attributed to differences in annotator ethnicity—in the Hate Speech detection datasets (DICES-350, DICES-990, Sap) this is the strongest contributor to polarization. In each case, statistically significant effects emerge for at least one of the major ethnic groups represented in the data (African Americanss (AAs) (-0.0428)/Asians (0.0659) for DICES-350, AAs (-0.2967) / Whites (0.1400) for Sap, Whites (-0.0377) for Kumar, and all ethnicities for DICES-990). This suggests that ethnic background constitutes a systematic source of annotator disagreement rather than isolated noise, and is more prevalent on tasks explicitly investigating racism.

Some dimensions exert an overall influence on the dataset

There is a general tendency for the *apunim* values of subgroups of the same PC to sum to zero. This indicates that overall, this PC does not contribute to the polarization of this task, or equivalently, that its subgroups "cancel each other out". Examples of this pattern can be found in Kumar for PCs `''Education''`, `''Gender''`, `''Political Affiliation''`, `''Religion Important''` and `''Technology Impact''`.

Some PC dimensions however do not necessarily sum to zero, which suggests that *these dimensions may exert a systematic influence* on the total polarization of the dataset. In other words, some annotator characteristics can consistently increase or decrease observed polarization (§3.3). In the Kumar dataset, for example, respondents who report having been previously targeted contribute positively to polarization ($apunim = 0.0349$), while those who have not been targeted contribute much more negatively ($apunim = -0.0633$). Similarly, transgender annotators contribute negatively to polarization ($apunim = -0.0894$) which is not balanced out by cisgender ones ($apunim = 0.0013$). These asymmetries indicate that some characteristics shape the overall polarization of the dataset rather than simply redistributing it across groups.

Less educated annotators are more subjective

Education tends to have a moderate but consistent effect across the Kumar and DICES-990 datasets; the more educated an annotator is, the more they fundamentally disagree with other annotators. In the Kumar dataset, for instance, Bachelor's-degree holders show a substantial positive contribution ($apunim = 0.0401$), while groups with lower education such as "College, no degree" exhibit a negative contribution ($apunim = -0.0335$). The same pattern can found in DICES-990, where annotators with at most a high-school education exhibit negative polarization attribution $apunim = -0.0751$, versus college-educated annotators whose values are extremely close to 0 (hence, why they are not shown in the Table—the full results can be found in E.2). This may indicate that less educated annotators are more likely to exhibit subjective views, while educated annotators will tend to broadly agree with each other.

5.3 Utilizing ordinal attributes

Analysis on annotated data frequently ignores the different kinds of measurements in studies using sociodemographic and LIKERT-style attributes. One could treat all dimensions as *nominal/categorical*, even if they are inherently ordinal (e.g., LIKERT-scale, education level) or even binarize when many values exist (e.g., age). With this approach, however, not only do we lose valuable information, but we may magnify statistical errors [20]. By contrast, we posit that *retaining the order of ordinal attributes reveals interesting trends* that would be otherwise missed with categorical inference.

Figure 6 shows *apunim* trends per ordinal PC attribute as shown in Table 2, focusing only on attributes with at least two statistically significant values ($p < 0.05$). This condition is necessary, since we can not infer trends from dimensions with a single non-zero *apunim* value. Note that `Age` is typically a ratio (positive, continuous) variable [20, 21], although it is treated either as ratio (Sap), interval (Kumar), or ordinal (DICES) variable depending on the dataset. Since ratio and interval variables are also ordinals, we treat `Age` as ordinal in this analysis.

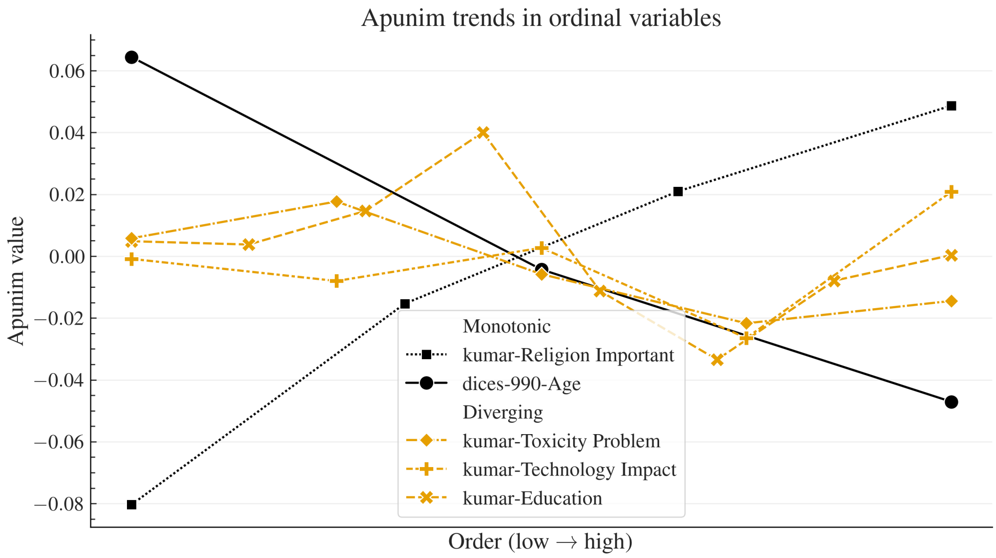

Fig. 6 Apunim values across ordinal factor levels with at least two statistically significant values. The x-axis corresponds to the ordered levels of each factor and is normalized so that different ordinal scales are comparable (e.g., education which is measured in different scales across datasets). The left side of the x-axis (low orders) refers to young annotators, low education, low religiousness, and indifference towards the impacts of toxicity and technology. The full ordinal modes for each dimension and their individual labels can be found in E.1.

We can see that `Religion Importance` is characterized by a strict monotonic increasing trend. This indicates that non-religious annotators (low order) are more divided among themselves, while very religious annotators (high order) agree among themselves, but fundamentally disagree with other annotators. A very similar, inverse trend is observed for `Age`. For `Education`, we observe a peak on the middle (4—"Bachelor's Degree"), which indicates two findings. First, up to an education level and as their level increases, people tend to agree more with people from a similar level. Second, beyond that level and till they reach the highest level, people tend to disagree more on their own level compared to ones from a different level.

6 How many annotators do we need?

While our test works best with many annotations per item, the cost required for multiple human annotators is often greatly restrictive [16]. Thus, it is imperative to find the least annotators needed for *apunim* to generate stable and reliable estimates.

For each item per dataset, we randomly sample with replacement a subset of annotators and compute the pol_{obs} value (see Equation 3). We repeat the procedure 30 times (since larger values lead to large computation times—see D), and compute the standard deviation. We removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of comments in DICES-990 were annotated by more than 73 annotators—see B.1). Figure 7 demonstrates the effect of

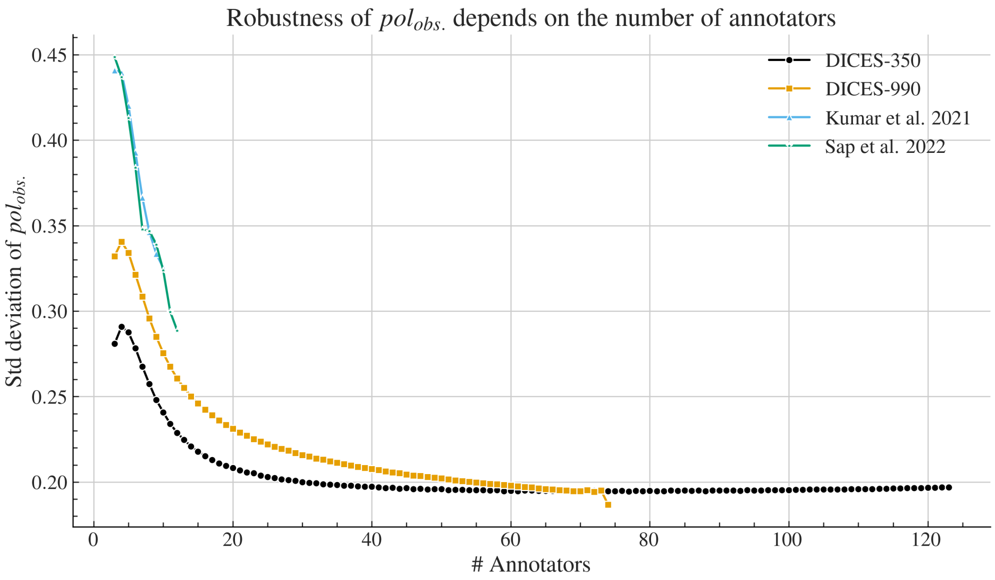

Fig. 7 Standard deviation of 30 randomly sampled (with replacement) pol_{obs} values (Eq. 3). The x-axis shows the sample size (number of annotators), starting from 3 and increasing one at a time, up to the largest annotator count for which a sufficient number of items exists (B.1).

the number of annotators on the reliability of the polarization estimation, which we discuss next.

Each dataset features a different standard deviation even with the same number of annotators, which is broadly consistent with the amount of polarization in each dataset (Appendix B.2). However, we observe a clear, non-linear, inverse relationship between the number of annotators and the standard error across datasets. This can be attributed to a more robust histogram estimation used by nDFU (§3.3) when provided with more points for each of the subgroups. After about 20 points, the estimation is good enough that more annotations contribute trivially to the robustness of the metric. Furthermore, we can see that the metric becomes consistent with about 20 annotators per comment for the DICES datasets, after which we start to get diminishing results. Unfortunately, more datasets with large numbers of annotators that include data about their PC characteristics, are needed to generalize this finding.

7 Conclusion

We introduced the *apunim* metric which attributes polarization to annotator subgroups for various PC dimensions. Our metric works well for imbalanced annotator splits, is well-defined for multi-label annotation, includes a statistical significance test, and unlike previous work, quantifies the amount of attribution in order to allow for comparisons between various experiments and datasets. Using this metric, we discover that annotator ethnicity is the strongest and most consistent driver of polarization between annotators across datasets, and is much more pronounced on Hate Speech detection tasks. We also discover that education and religion consistently impact the behaviour of annotators. Specifically, we discover that very religious annotators fundamentally agree with themselves but disagree with others in Toxicity detection tasks, a trend that is gradually and consistently diminished, and ultimately reversed, by irreligious annotators. We also provide an open-source, PyPi-installable implementation, and an estimation of the number of annotators needed for each item, for our method to be consistently reliable.

8 Discussion

Future work

Our framework has broad applications in NLP sub-tasks relating to content moderation. Since many such tasks are run on multiple discussions, our framework can be used on the discussion, instead of the dataset level. In that case, researchers may be able to isolate polarizing discussions, and investigate underlying patterns of polarization for individual minority groups. Alternatively, items can be divided by comment turn, allowing researchers to investigate the evolution of polarization across discussions according to minority groups. Lastly, our work can be used to more reliably test whether LLM-as-a-judge systems can replicate the polarization patterns of certain minority groups.

Limitations

Since the predominant paradigm in Machine Learning is the use of "gold-label" datasets, researchers often only publish single-label, aggregated data. Even when they do not, these datasets rarely contain metadata on annotator PC characteristics. Our analysis and the applicability of our framework is thus constrained by the lack of availability of such datasets. Furthermore, the substantial costs of human annotation, prevent us from applying this framework to an item-level analysis.

AI use Statement

LLMs were used for minor styling changes in this document, for some implementation details in the Python framework, as well for some of the code used for figure generation. We used both manual and automated supervision (unit tests) for the code generation in the framework. The authors have verified all other LLM outputs.

Acknowledgments

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Appendix A Acronyms

LLM Large Language Model
PC Personal Characteristic
nDFU normalized Distance From Unimodality
FWER Family-Wise Error Rate
NLP Natural Language Processing
GPLv3 GNU General Public Licence v3
AA African Americans

Appendix B Annotation details

B.1 Annotator distribution

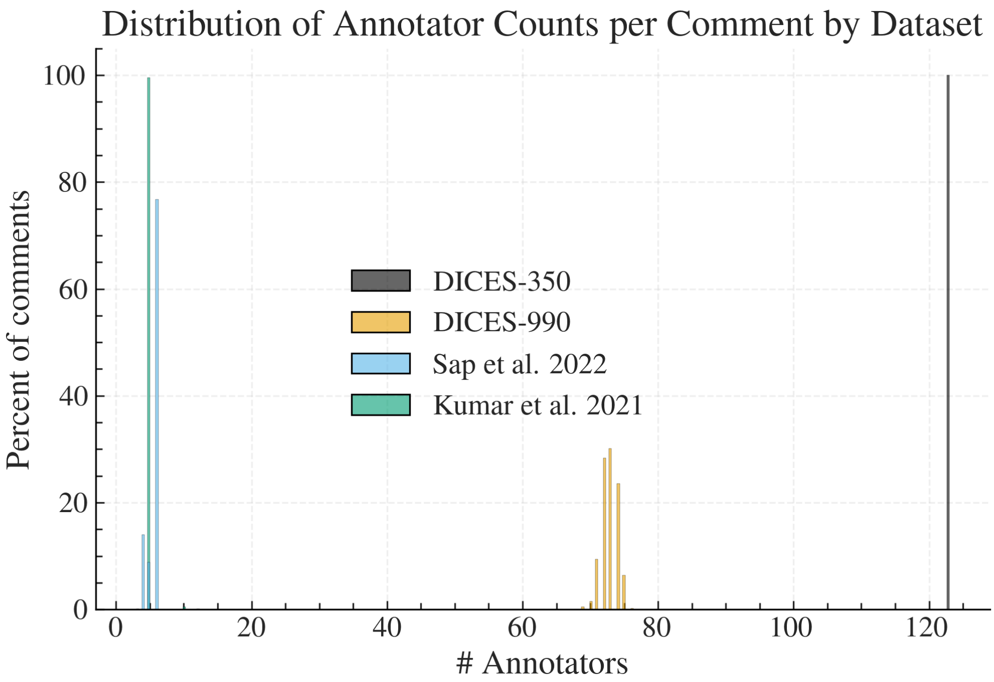

Fig. B1 We calculate the ratio of comments for each dataset (y-axis) that had x number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few comments in the Kumar dataset exceeding this threshold (see Table B1).

The number of annotators per comment for all datasets presented in §4 can be found in Figure B1, and descriptive statistics can be found in Table B1. The DICES-990 dataset has exactly 123 annotations per comment, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample

Table B1 Descriptive statistics for the number of annotations per comment, grouped by dataset.

	count	mean	std	min	25%	50%	75%	max
DICES-350	350	123	0	123	123	123	123	123
DICES-990	990	72.8313	1.1652	69	72	73	74	76
Kumar et al. 2021	20000	5.0925	4.8872	5	5	5	5	650
Sap et al. 2022	585	5.6359	0.7713	3	6	6	6	12

used from the Kumar dataset exhibits a few comments that vastly exceed the promised 5 number of annotators. We included these comments for the experiment presented in Table 2, but not for the annotator number experiment in Figure 7 since they would produce uninterpretable figures.

B.2 Polarization

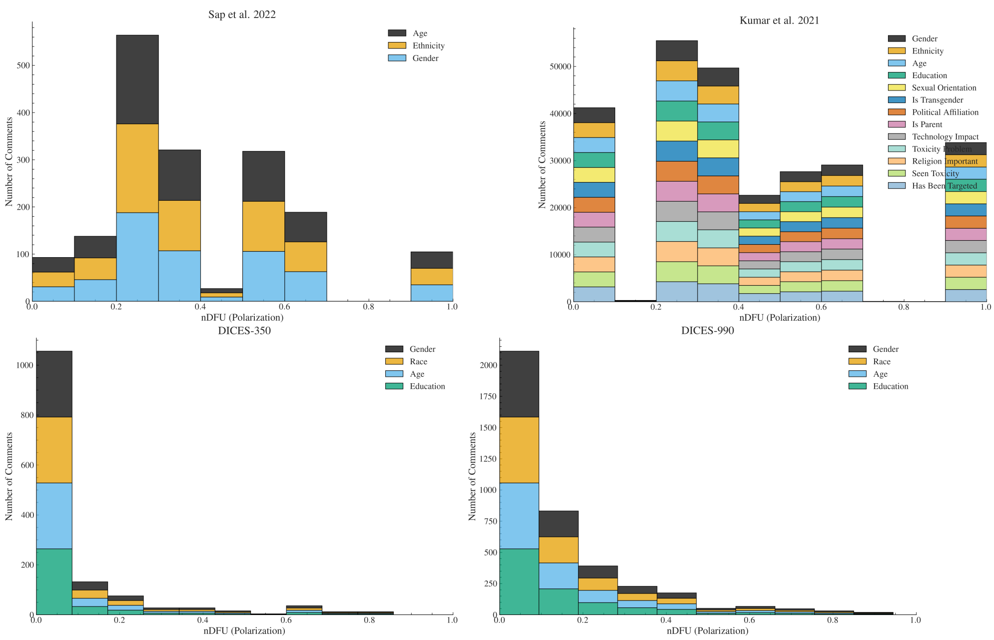

Fig. B2 Histogram of dataset polarization for the human datasets considered in this paper per PC.

Appendix C Software

C.1 Installation and usage

We release the `apunim` package, which is a Python library that implements both the nDFU and apunim calculations, as well as its statistical test. The software is available on PyPi (https://pypi.org/project/apunim). We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (https://dimits-ts.github.io/apunim).

The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project's repository (https://github.com/dimits-ts/apunim). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy, scipy, statsmodels`).

C.2 Implementation details

By default, the computation of the apunim values happens on a single thread. In cases of serious computational cost, we recommend running the apunim calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple PC dimensions, each of which is computationally expensive.

In order to lower the computational cost required, we re-use the random partitions $\tilde{P}_i$), which are originally calculated to obtain the apriori polarization values (see Equation 2), for the p-value calculations (§3.4). This means that in our current implementation, the number of groups for estimating apriori polarization ($\tilde{P}_i$ in Equation 2), and the number of apunim values used in the p-value calculation ($\tilde{D}_i$ in Algorithm 1) have to be the same.

Appendix D Replication Details

We use version `1.0.1` of the `apunim` library for our experiments. We use 100 random iterations for the apriori value calculation (Equation 2) and the p-value estimation (§3.4), and set a FWER rate of 0.95.

The current version of the library handles large number of annotators very well due to the use of vectorized operations, but struggles when faced with a lot of comments when the annotators belong to multiple groups. Indeed, while the experiments for all datasets presented in §4 need no more than a minute to run, the Kumar dataset requires about 12 hours of computation even with a 20% sample. The annotator variance analysis presented in §6 also required more than 10 hours of computation, although that is expected due to the nature of the experiment, which is very computationally demanding, and a non-standard use of our library.

This bottleneck is likely caused by repeated creation of numpy arrays, resulting in repeated memory allocations for each subgroup, in each comment, for each PC dimension. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each PC subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` linux package [22] for our own experiments).

Appendix E Full Results

E.1 Ordinal dimension analysis

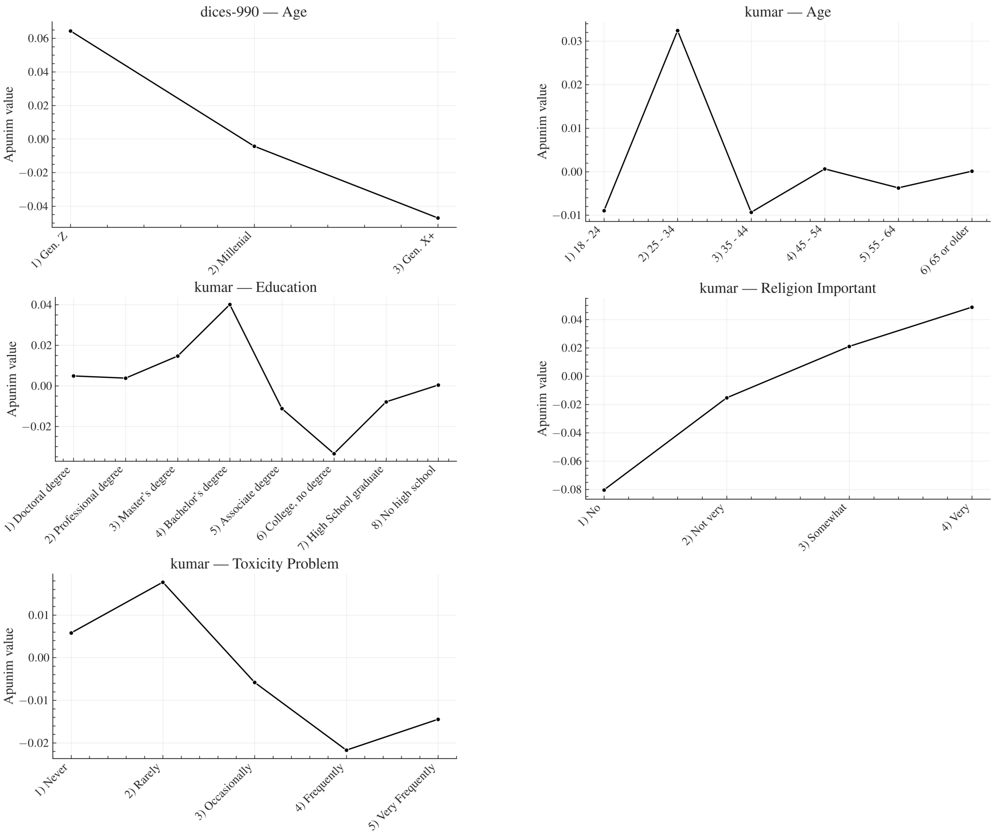

Fig. E3 Apunim values across ordinal factor levels with at least two statistically significant values, expanded for each ordinal PC. This figure is complementary to Figure 6. The values in the graphs were extracted from the Tables of E.2.

E.2 Apunim results

Table E2 Aposteriori unimodality results for the DICES-350 dataset.

PC dimension	Value	apunim	support
Age	1) Gen. X+	0.0186	9703
	2) Millenial	-0.0059	11268
	3) Gen. Z	-0.0066	17528
Education	College or higher	0.0124	23475
	High school or lower	-0.0152	12833
	Other	-0.0214*	2191
Gender	Man	-0.0193	19093
	Woman	0.0215	19406
Race	African Am.	-0.0428***	9077
	Asian	0.0659***	8138
	Latino	-0.0030	6886
	Multiracial	-0.0001	5008
	White	-0.0198	9390

Table E3 Aposteriori unimodality results for the DICES-990 dataset.

PC dimension	Value	apunim	support
Age	1) Gen. X+	-0.0471***	14384
	2) Millenial	-0.0043	38436
	3) Gen. Z	0.0644***	13741
Education	College or higher	0.0095	59142
	High school or lower	-0.0751***	7419
Gender	Man	0.0278***	32494
	Woman	-0.0235***	33471
Race	African Am.	0.0909***	5810
	Latino	-0.0030	6886
	Multiracial	-0.0001	5008
	White	-0.0198	9390

Table E5: Aposteriori unimodality results for the Kumar dataset.

PC dimension	Value	apunim	support
Age	1) 18 - 24	-0.0090	9742
	2) 25 - 34	0.0324***	33203
	3) 35 - 44	-0.0093	21055
	4) 45 - 54	0.0006	11025
	5) 55 - 64	-0.0037	6123
	6) 65 or older	0.0001	2567

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Table E5: Aposteriori unimodality results for the kumar dataset.

PC dimension	Value	apunim	support
Education	1) Doctoral degree	0.0048	1019
	2) Professional degree	0.0038	1321
	3) Master's degree	0.0147**	13420
	4) Bachelor's degree	0.0401***	34085
	5) Associate degree	-0.0113**	9122
	6) College, no degree	-0.0335***	16666
	7) High School graduate	-0.0079*	7326
	8) No high school	0.0004	462
Ethnicity	Asian	-0.0019	5062
	Black	0.0110*	10999
	Hispanic	-0.0025	2389
	Multiracial	-0.0041	3682
	Other	0.0022	1507
	Unknown	-0.0030	1160
	White	-0.0377***	45306
Gender	Female	-0.0278***	40686
	Male	0.0275***	37929
	Nonbinary	-0.0008	489
	Unknown	-0.0029	621
Has Been Targeted	False	-0.0633***	45234
	True	0.0349***	24491
Is Parent	No	-0.0707***	37231
	Prefer not to say	-0.0023	831
	Yes	0.0535***	41493
Is Transgender	No	-0.0894***	11978
	Prefer not to say	0.0005	856
	Yes	0.0013	2306
Political Affiliation	Conservative	0.0331***	23010
	Independent	-0.0157***	22818
	Liberal	-0.0287***	32917
	Other	-0.0049	1822
	Prefer not to say	0.0015	3548
Religion Important	1) No	-0.0804***	26008
	2) Not very	-0.0154***	10055
	3) Somewhat	0.0210***	20169
	4) Very	0.0487***	27284
	Prefer not to say	0.0001	1179
Seen Toxicity	False	0.0311***	20185
	True	-0.0717***	43265

Continued on next page

Table E5: Aposteriori unimodality results for the kumar dataset.

PC dimension	Value	apunim	support
Sexual Orientation	Bisexual	0.0175***	10005
	Heterosexual	-0.0571***	38048
	Homosexual	-0.0017	2975
	Other	-0.0003	728
	Prefer not to say	-0.0017	1751
Technology Impact	1) Very negative	-0.0009	784
	2) Somewhat negative	-0.0080	7973
	3) Neutral	0.0027	10720
	4) Somewhat positive	-0.0265***	40477
	5) Very positive	0.0208***	22471
Toxicity Problem	1) Never	0.0058	4962
	2) Rarely	0.0177***	14457
	3) Occasionally	-0.0058	29476
	4) Frequently	-0.0217***	24774
	5) Very Frequently	-0.0145***	10656

Table E4 Aposteriori unimodality results for the Sap dataset.

PC dimension	Value	apunim	support
Age	1) Gen. X+	0.0192	743
	2) Millennial	0.0189	1792
	3) Gen. Z	-0.0153	239
Ethnicity	Black	-0.2967***	1041
	White	0.1400***	2048
Gender	Male	0.0942***	1635
	Female	-0.1810***	1333